METRIC

DOD-B-63545
19 FEBRUARY 1982

MILITARY SPECIFICATION

BOLTS, RIBBED SHOULDER, WHEEL MOUNTING, METRIC
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of wheel mounting bolts with ribbed shoulder, metric dimensions.

1.2 Classification. Bolts shall be of the following types as specified on the applicable specification sheets (see 6.2).

Type I. - Bolt, ribbed shoulder, wheel mounting
Type II. - Bolt, ribbed shoulder, extended shoulder, wheel mounting

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
QQ-Z-325 - Zinc Coating (Electrodeposited)
PPP-H-1581 - Hardware (Fasteners and Releated Items), Packaging and Packing for Shipment and Storage of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5306

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.

MILITARY

DOD-B-63545/1 - Bolts, Ribbed Shoulder, Wheel Mounting, Metric
DOD-B-63545/2 - Bolt, Ribbed Shoulder, Extended Shoulder, Wheel Mounting, Metric

STANDARDS

FEDERAL

FED-STD-H28/21 - Screw Thread Standards for Federal Services, Section 21, Metric Screw Threads
FED-STD-66 - Steel, Chemical Composition and Hardenability

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-1312 - Fasteners Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J 1102, Mechanical and Material Requirements for Wheel Bolts

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 19096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.

3.2 Material. There is no exclusion to the use of recovered materials.

3.2.1 Alloy steel. Bolts shall be made of alloy steel conforming to FED-STD-66, as indicated on specification sheets.

3.3 Design.

3.3.1 Threads. Threads shall conform to the "M" profile and tolerance class 6H, after plating, in accordance with FED-STD-H28/21. Threads shall be formed by rolling process.

3.3.2 Head taper. Bolt head taper (angle between any side of the head and the bolt axis) shall be not greater than 2 degrees.

3.3.3 Bearing surface. The bearing surface of the bolt head shall be perpendicular to the bolt axis within 2 degrees for shank lengths up to 100 mm, and 1 degree for shank lengths over 100 mm.

3.4 Mechanical properties. Mechanical properties of bolts shall be as specified in table I.

TABLE I. Mechanical properties.

Nominal size (M thread)	Minimum area (mm^2)		Shear strength (kN) min		Ultimate tensile 1/ load (kN) min
	Thread	Shoulder	Single 2/	Double 3/	
12	84.3	119.4	52.27	148.05	87.17
20	245.0	397.6	151.90	493.02	253.33
22	303.0	471.4	187.86	584.53	313.30

1/ Ultimate tensile strength of the bolt is based on a minimum ultimate tensile strength of 1034 MPa for the material and the minimum thread area.
2/ Single shear strength is based on a stress of 60% of the minimum ultimate tensile strength 620 MPa and the minimum thread area.
3/ Double shear strength is based on a stress of 60% of the minimum ultimate tensile strength 620 MPa and twice the minimum shoulder area.

3.4.1 Hardness. Bolts shall have a surface hardness and core hardness as indicated on specification sheets.

3.5 Protective finish. Bolts shall be cadmium plated in accordance with QQ-P-416, type II, class 1, 13 µm thick or zinc coated in accordance with QQ-Z-325, type II, class 2, 13 µm thick as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.

3.6 Workmanship. Finished bolts shall be uniform in quality and appearance. Surfaces shall be free of foreign matter. Bolt surfaces in the unplated condition shall be free of cracks, bursts, folds, tool marks, seams, voids on the bearing surface, nicks and gouges.

3.7 Identification marking. Bolts shall be marked in accordance with MIL-STD-130.

3.8 Serrations. Bolt serrations shall be uniform in quality, quantity and appearance. In the unplated condition they shall be free of cracks, voids, nicks, and gouges. Serrations shall not strip when tested in accordance with paragraph 4.5.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that materials used in fabricating bolts are in accordance with 3.2, 3.4 and 3.5.

4.3.1 Protective finish. Sample bolts shall be inspected for minimum thickness and continuity of the plating in accordance with QQ-P-416, or QQ-Z-325.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Group A Examination	1/	4.5.1
Group B Mechanical properties and serrations	3.4 3.8	4.5.2 4.5.3

1/ All requirements not covered by tests.

4.4.1 Inspection lot. An inspection lot shall consist of all bolts covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.4.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.3 Sampling for group A inspection. Sampling shall be in accordance with MIL-STD-105 inspection level II, with acceptable quality level (AQL) of 1.5 percent defective for major defects. For minor defects, inspection level I with an AQL of 4.0 percent defective shall apply.

4.4.3.1 Classification of defects. Defects shall be classified as specified in table III.

TABLE III. Classification of defects.

Category	Defect	Inspection method
Critical	None defined	
Major		
101	Threads, not as specified (see 3.3.1)	Gage
102	Shoulder diameter, not as specified (see 3.1)	SIE 1/

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects. - Continued

Category	Defect	Inspection method
Minor		
201	Overall length of bolt, not as specified (see 3.1)	SIE
202	Shoulder length, not as specified (see 3.1)	SIE
203	Other dimensions, not as specified (see 3.1)	SIE
204	Protective finish missing (see 3.5)	Visual
205	Head taper, not as specified (see 3.3.2)	SIE
206	Bearing surface-bolt axis perpendicularity, not as specified (see 3.3.3)	SIE
207	Workmanship, not as specified (see 3.6)	Visual
208	Packaging, not as specified (see 5.1)	Visual

1/ Standard inspection equipment

4.4.4 Sampling for group B inspection. Sampling shall be in accordance with MIL-STD-105.

4.4.4.1 Disposition of sample units. Bolts which have been subjected to group B inspection shall be delivered on the contract.

4.4.5 Packaging inspection. The sampling and inspection of the preservation-packaging, packing, and container marking shall be in accordance with PPP-H-1581.

4.5 Methods of inspection.

4.5.1 Examination. Sample bolts shall be thoroughly examined to determine conformance with this specification and applicable specification sheets with respect to all the requirements not covered by tests.

4.5.2 Mechanical properties. Sample bolts shall be tested for conformance to requirements of 3.4. Tensile strength, double shear, and single shear strength shall be determined in accordance with MIL-STD-1312, test methods 8, 13, and 20, respectively. Surface hardness shall be determined in accordance with MIL-STD-1312, test method 6.

4.5.3 Serration test. Sample bolts shall be subjected to serration test in accordance with SAE J 1102, Serration test.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Bolts covered by this specification are intended to be used primarily for mounting automotive vehicle wheels.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specificatiion and the applicable specification sheet.
b. Applicable specification sheet part number (see 3.1).
c. Level (degree) of protection, in accordance with PPP-H-1581, ordering data (see 5.1).

Custodians:
Army - AR
Navy - AS
Air Force - 99

Preparing activity:
Army - AR

(Project 5306-0692)

Review activities:
Army - AT
Navy - SH, MC
DLA - IS
NSA - NS

User activities:
Army - ME
Navy - OS

Agent:
DLA - IS

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE. This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE DOD-B-63545
BOLTS, RIBBED SHOULDER, WHEEL MOUNTING, METRIC GENERAL SPECIFICATION FOR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATI

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Research and
Development Command
ATTN: DRDAR-TST-S
Dover, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:04Z
Check the source to verify that this is the current version before use.